

B.A./B.Sc. (Hons.) Semester V Examination 2022-23
STATISTICS

STB-501/STA-501: Applied Statistics

Time: Three hours

Max. Marks: 70

Note: Answer any FIVE questions. All questions carry equal value.

1. (a) What are the standard methods of obtaining vital statistics? Detail the key administrative and demographic uses of vital statistics data.
(b) Define Gross Reproduction Rate (GRR) and Net Reproduction Rate (NRR). Explain how far they can be considered as indices of population growth, and interpret the following joint scenarios:
 - i. $GRR = 2.10$ and $NRR = 1.10$
 - ii. $GRR = 1.15$ and $NRR = 1.10$
 - iii. $GRR = 1.75$ and $NRR = 1.80$
2. (a) Discuss the mathematical framework of the logistic curve used for modeling and measuring the growth of biological populations.
(b) What do you mean by a life table? State clearly the fundamental assumptions underlying a standard life table. Describe the definitions and functions of its various columns, and outline its primary applications.
3. (a) Define Crude Death Rate (CDR) and point out its core limitations. Explain clearly the concept of standardized death rates, including both the direct and indirect methods of their computation.
(b) Describe different measures of fertility (including CBR, GFR, ASFR, and TFR) along with their respective merits and demerits.
4. (a) Discuss the economic importance, applications, and structural limitations of index numbers.
(b) What is the logical need and significance of conducting a family budget enquiry in the construction of a cost of living index number?
(c) Define the time reversal test for index numbers. Mathematically prove that Fisher's ideal index number satisfies the time reversal test.
5. What is industrial quality control? Describe the main underlying principles of statistical quality control (SQC) charts. Discuss the operational need and utility of SQC in manufacturing setups, and describe the detailed construction of control charts for the mean ($\bar{X}$) and the standard deviation (σ).
6. (a) Discuss in detail the underlying distribution, statistical foundations, and construction of fraction defective (p) charts and number of defects (c) charts.
(b) Define the Infant Mortality Rate (IMR). Why do we have to calculate adjusted infant mortality rates? Describe in detail Logan's method of adjustment for infant mortality rates.
7. (a) Explain in detail the concept of a secular trend in a time series. Critically examine the various analytical and mathematical methods used for measuring trends.
(b) Write short notes on the different structural components of a time series. Using clear real-life examples, explain the core differences between seasonal and cyclical variations.

8. (a) What do you mean by a time series? Describe its practical and forecasting uses in business and economic data analysis.
- (b) Explain the classical additive and multiplicative decomposition models of a time series, stating clearly the assumptions behind both frameworks.
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ExternalPortals&AcademicResources

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app/>